AP PRECALCULUS

Unit 1 • Worksheet 1.5A

# Worksheet 1.5A • Polynomial Functions and Complex Zeros

Name: _____ Date: _____

**Directions**

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

### Question 1

NO CALCULATOR

What are the zeros of $p(x) = x^2(x-4)(x+1)$?

(A) $0,\ 4,\ -1$
(B) $0,\ -4,\ 1$
(C) $4,\ -1$
(D) $2,\ 4,\ -1$

### Question 2

NO CALCULATOR

$p(3) = 0$ for a polynomial $p$. Which must be true?

(A) $(x+3)$ is a factor of $p(x)$.
(B) $(x-3)$ is a factor of $p(x)$.
(C) $p(0) = 3$.
(D) $p$ has degree 3.

### Question 3

NO CALCULATOR

Which polynomial has $x$-intercepts at $-2$, 1 and 5 only?

(A) $(x-2)(x+1)(x+5)$
(B) $(x+2)(x-1)(x-5)$
(C) $(x+2)(x-1)(x+5)$
(D) $x(x+2)(x-1)(x-5)$

### Question 4 (Free response)

NO CALCULATOR

Find all real zeros of each polynomial.

(a) $f(x) = x^3 - 9x$ (b) $g(x) = x^3 - 2x^2 - 8x$
(c) $h(x) = x^4 - 13x^2 + 36$ (d) $k(x) = 2x^2 - x - 3$